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Soil properties and maple–beech regeneration a decade after liming in a northern hardwood stand

Jean-David Moore *, Louis Duchesne, Rock Ouimet

Direction de la recherche forestière, Ministère des Ressources naturelles et de la Faune du Québec, 2700 rue Einstein, Québec, Canada G1P 3W8

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ABSTRACT

Soil properties, regeneration, sugar maple (SM, *Acer saccharum* Marsh.) crown dieback, and light interception were evaluated following dolomitic lime application ($CaMg(CO_3)_2$; 0, 2, 5, 10 and 20 t ha^{-1}) in a base-poor and declining northern hardwood stand of Quebec (Canada). Ten years after liming, soil pH, cation exchange capacity and base saturation increased with the lime rate. Meanwhile, concentrations of exchangeable K, Na, and acidity (0–20 cm) decreased. Concentrations and content of total carbon, organic matter, and total N decreased in the top sampled soil layer, mainly for the highest lime rate. Mean crown dieback of SM ranked from 0 to 3.4% among lime treatments, while it ranked 18.4% for the unlimed controls. Light interception by the canopy responded with an opposite pattern to crown dieback, with higher interception in lime treatments and lower in control. Proportion and diameter of SM regeneration stems increased with lime rate while proportion of American beech (AB, *Fagus grandifolia* Ehrh.) regeneration stems decreased in the understory. Height and stem diameter of the three taller regeneration stems (mainly AB) in each plot were inversely correlated with lime rate. Overall, the results suggest that the increase in soil fertility following liming had a beneficial effect on SM regeneration within 10 years, even if light availability was lower under treated SM than under control, and had an overall negative impact on AB regeneration.

1. Introduction

Sugar maple (SM; *Acer saccharum* Marsh.) has a broad distribution in North America, and thrives throughout the northeastern United States and southeastern regions of Canada. Many studies in the Northeast have demonstrated the importance of calcium (Ca) for SM vitality (Heisey, 1995; Ouimet and Camiré, 1995; Wilmot et al., 1995, 1996; Sharpe et al., 1999; Horsley et al., 2000, 2002; Moore et al., 2000; Duchesne et al., 2002; Moore and Ouimet, 2006). Calcium, the fifth most abundant element in trees, is an important component for wood formation and maintenance of cell wall (e.g. Lawrence et al., 1995). This nutrient also plays a role in fine-root growth of SM (Adams and Hutchinson, 1992). Moreover, some studies suggest better photosynthetic rates for SM with increasing Ca availability (Ellsworth and Liu, 1994; Wilmot et al., 1996). In recent decades, however, acid deposition increased leaching losses from soil of base cations, particularly Ca, above the replenishment rate by chemical weathering and atmospheric depositions, causing a reduction in the availability of mineral nutrients in some non-calcareous soils (Bailey et al., 2005; Houle et al., 1997; Likens et al., 1998; Ouimet et al., 2001, 2006; Watmough and Dillon, 2003).

Although other hypotheses were developed to explain the failure of SM regeneration and the large increase of pole-size American beech (AB; *Fagus grandifolia* Ehrh.) trees observed in northeastern North America in last decades, such as the indirect effect of beech bark disease (Hane et al., 2003), recent studies suggest that acid deposition and subsequent depletion of base cations and acidification of soils, combined with SM dieback in some areas, are another likely explanation to this phenomenon (Jenkins, 1997; Duchesne et al., 2005). Moreover, the absence of AB growth response to liming noted in the Long et al. (1997) study, contrary to SM, suggests that AB is not highly sensitive to the acid–base status of soils. This reasoning seriously questions the sustainability of SM in Ca-deficient and declining northern hardwood stands, and suggests that the importance of AB in many regions could increase at the expense of SM if no intervention is taken regarding Ca deficiency. It has been demonstrated that soil exchangeable Ca depletion and its influence on seedling dynamics could lead to substantial decreases in SM

* Corresponding author. Tel.: +1 418 643 7994x6529; fax: +1 418 643 2165.
E-mail address: jean-david.moore@mrnf.gouv.qc.ca (J.-D. Moore).

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canopy dominance within a single forest generation (<125 years, Kobe et al., 2002).

To restore soil Ca status and SM tree health, Ca addition has been tested in Ca-poor northern hardwood soils (Wilmot et al., 1996; Long et al., 1997; Moore et al., 2000; Moore and Ouimet, 2006; Juice et al., 2006). Although beneficial effects of Ca treatment on SM trees were demonstrated in these studies, only a few of them have evaluated the effect of this treatment on SM regeneration (Long et al., 1999; Kobe et al., 2002; Juice et al., 2006; Bigelow and Canham, 2007). In Pennsylvania, Long et al. (1999) observed that SM seedling survival was higher (70%) in limed (22.4 t ha^{-1}) compared with unlimed plots (32%), 5 years after treatment. In New Hampshire, Kobe et al. (2002) observed a 52% increase in relative diameter growth of SM seedlings in Ca-amended plots (100 kg Ca ha^{-1}) compared to controls, after 2 years. In the same area, Juice et al. (2006) also reported a much higher survivorship of SM seedlings on Ca-treated (36.6%) than on a control watershed (10.2%), 5 years after treatment.

Given the importance of regeneration for future forest composition and structure, we evaluated the mid-term (11 years) effect of liming on regeneration in a Ca-poor northern hardwood stand. We hypothesized that liming changed soil properties and that these changes caused a beneficial effect on SM regeneration abundance and growth, but no effect on AB regeneration.

2. Materials and methods

2.1. Site description

The Lake Clair Watershed (46′57″, 71′40″, 285 m a.s.l.) is located approximately 50 km northwest of Quebec City, QC. The forest stand is uneven-aged and dominated by SM in association with AB and yellow birch (*Betula alleghaniensis* Britt.), with basal areas of 17.3, 7.3, and 4.0 m^2 ha^{-1}, respectively (2006 survey). Dominant and codominant SM trees are between 85 and 130 years old with an average height and diameter at breast height (DBH) of 20.2 m and 27.6 cm, respectively. Dendrochronological analysis revealed that basal area increment of dominant and co-dominant SM trees followed a negative trend over the last 40 years which is a strong indicator of a true tree growth decline (Duchesne et al., 2002, 2003). Forest inventory data revealed a 20% decrease of SM while AB increased by 128% in terms of basal area over the last 20 years. No sign of beech bark disease has been reported in this area. The soil is a well-drained Typic Haplorthod (Soil Survey Staff, 1998) or Orthic Ferro-Humic Podzol (Canada Soil Survey Committee, 1998) with a moder humus. Forest floor and soil pH in this area ranged from 2.8 to 3.7 and from 3.3 to 4.5, respectively (Houle et al., 1997, 2002; Moore, unpublished data). The site has been receiving atmospheric acid deposition in the order of 15 kg S ha^{-1} yr^{-1} and more since the 1970s and is being depleted in Ca (Houle et al., 1997). No earthworms were observed in this soil. Additional information on site and stand characteristics are provided in Houle et al. (1997, 1999).

2.2. Experimental setup

In 1994, powdered and pelletized dolomitic lime was applied on forest soils in an experimental forest section located at the border of the Lake Clair Watershed. This liming experiment was established to determine the nutrient, vigor and growth response of SM to amendments (Moore et al., 2000; Moore and Ouimet, 2006). Briefly, 98 SM, spaced by at least 15 m one from the other, were selected and randomly distributed among eight $CaMg(CO_3)_2$ treatments (0, 0.5, 1, 2, 5, 10, 20, 50 t ha^{-1}). The 1 t lime ha^{-1} rate corresponded approximately to the current total exchangeable Ca in the soil (200 kg ha^{-1}, Houle et al., 1997). Dolomitic lime was applied evenly within a 5-m radius around each tree in the last week of August 1994, after cutting and removing the entire understory over the same radius to reduce variability of understory abundance and to ensure uniform spreading of lime. No other tree stem was in the 5-m radius around SM trees. For the purpose of the present study, five trees were randomly selected from the 0, 2, 5, 10 and 20 t ha^{-1} treatments, for a total of 25 trees.

2.3. Regeneration sampling

Two inventory types were carried out in this study. First, in August 2005, regeneration stems were numbered and their diameter (at 15 mm from the surface of the forest floor) and height were measured in four circular sub-plots of 0.20 m^2 (radius 25 cm) located on cardinal points at 2.5 m from the center of the stem, for a total of 0.80 m^2 per plot, representing 1% of the plot area. Systematic sub-plots distribution for sampling an entire plot, as made in our study, is often considered more important than sample size (Cochran, 1977). Also, the surface sampling area was comparable to other published surveys on the subject (Long et al., 1999: 1.8%; Juice et al., 2006: 0.07%). The portion of sub-plot occupied by rocks or woody debris was visually estimated and subtracted from the total plot area. Second, height and stem DBH at 15 cm from the surface of the forest floor of the three taller regeneration stems were measured in each 5-m radius plot. Some observations made along the 11-year study confirmed that probably most of the taller regeneration stems of AB were sprouts. However, while it is likely that the lower regeneration stems of SM were mostly germinants, the origin of the taller regeneration stems of SM and the rest of AB regeneration was not determined.

2.4. Soil sampling and analysis

Ten years after liming, soils were sampled at two different locations around each tree. The first soil surface samples (0–12 cm) were extracted using a 261 cm^3 volumetric hammer soil sampler. Care was taken not to compact the soil during sampling. This first sample contained the forest floor (7–10 cm in thickness), a small eluviated horizon when present, and the top of the first B horizon. Volumetric sampling was carried out for this layer to ascertain whether soil organic C accumulated or not, because it was observed that organic C concentrations increased in the top soil layers with the lower liming rates 5 years after treatment (Houle et al., 2002). The mineral soil at 12–20 and 20–40 cm depth was qualitatively sampled separately using an auger. Samples of the same depth were pooled for each tree, air dried, weighed, and sieved to 2 mm before chemical analysis. Soil pH was measured using a soil-to-water ratio of 1:2.5. Exchangeable cations (K, Ca, Mg, Na, Al, Fe, and Mn) were extracted with an unbuffered NH_4Cl (1 M, 12 h) solution, and measured with an inductively coupled plasma emission spectrophotometer (Thermo Jarrell Ash ICAP-9000). Exchangeable hydrogen ions were directly measured in the extracts using a pH probe. Exchangeable acidity was estimated by calculating the sum of extracted H, Al, Fe, and Mn, assuming complete ion hydrolysis. Subsamples were ground to 0.5 mm for analysis of total carbon (dry combustion), organic matter (loss on ignition) and nitrogen (Kjeldahl) (Page et al., 1982).

2.5. Crown dieback and light measurements

In August 2004, crown dieback of SM trees was visually estimated (Moore and Ouimet, 2006). The percentage of light interception by the canopy, measured in August 2006, was also used as an index of crown defoliation and transparency, and to describe the resulting light environment under each tree. Above-canopy photosynthetic photon flux density 400–700 nm (PPFD) was measured with a point quantum sensor (Li-190, LICOR, Lincoln, NE), installed 1.5 m above the ground in an open area located approximately 250 m from the study site. This sensor was linked to a datalogger (CR-50, Campbell Scientific Inc.), which recorded readings taken every 10 s. A line quantum sensor (Li-191, LICOR, Lincoln, NE) was used to measure PPFD under the trees, 2 m above the ground. Two measurements were made above each of the four regeneration plots, for a total of eight measurements under each tree. Measurements were recorded in clear sky conditions between 11:00 and 13:00 eastern standard time, simultaneously with the clock of the sensor that was located in the open area. The two sensor measurements recorded in the open area were also compared. Values of measurements taken under the trees were then adjusted by linear regression (r^2 = 0.992) to take into account differences among the two sensors. Finally, the percentage of light interception under each tree was calculated by expressing the difference between the open area PPFD and the average PPFD measurement under trees as a percentage of open area PPFD.

2.6. Statistical analyses

The effects of lime treatment, species (SM, AB, and others), and their interaction on the abundance, height and diameter of regeneration were examined with analysis of variance (ANOVA) using the general linear models procedure (PROC GLM) in SAS (SAS Institute, 2002). The effects of lime treatments on the average height and diameter of the tree taller regeneration stems and on soils at the three depths were also examined. Linear and second degree polynomial contrasts were built to assess the effect of lime treatments. Calculation of the linear and quadratic coefficients for unequally spaced time intervals was made according to Robson (1959). When polynomial contrasts were not significants, but a step function was visually detected, additional specific contrasts were tested. Homogeneity of variance was tested by visual examination of the residuals. When heteroscedasticity was found, the data were transformed prior to ANOVAs.

3. Results

3.1. Soil

Ten years after the lime treatments, soil properties changed with the lime application rate, at least down to the deepest soil layer sampled (20–40 cm; Table 1). In the first 12 cm of soil, pH increased and exchangeable acidity decreased in a linear fashion with the lime rate, while concentrations of exchangeable Ca and Mg, cation exchange capacity (CEC) and base saturation (BS) increased in a quadratic fashion, mainly because of the 20 t ha^{-1} lime rate ($p \leq 0.014$). Soil pH and BS increased from (mean $\pm$ S.E.) 4.10 ± 0.09 and $41.9 \pm 5.1\%$, respectively, for the controls, to 5.51 ± 0.10 and $97.5 \pm 5.3\%$, respectively, for the 20 t ha^{-1} lime treatment. In contrast, concentrations of total carbon, organic matter, total N and exchangeable K and Na decreased in a quadratic fashion with the lime rate in this soil layer, this effect being mainly due again to the 20 t ha^{-1} lime rate ($p \leq 0.018$). Total carbon, total N and organic matter concentrations decreased by 38% for the 20 t ha^{-1} lime rate relative to the controls. The increase in exchangeable Ca and Mg and decrease in exchangeable K and acidity with the lime rate were also apparent in terms of content (mol(+) m^{-2}) in this soil layer (linear effect of lime rate; $p \leq 0.001$; Table 2). Reductions in total carbon, total N, and organic matter concentrations with the 20 t ha^{-1} lime rate in this soil layer were not associated with a statistically detectable change in its C/N ratio ($p = 0.668$; Table 1). Although these parameters did not change along with the lime rate in the first 12 cm when expressed as contents ($p \geq 0.233$; Table 2), comparison of the 20 t ha^{-1} lime rate with the other doses showed that the 15% decrease in total carbon ($p \leq 0.125$), total N ($p \leq 0.035$), and organic matter ($p \leq 0.040$) content, and the 24% increase in its soil bulk density ($p \leq 0.034$) were mostly significant (Table 2).

In the 12–20 cm soil depth layer, pH, exchangeable Ca and Mg, and BS increased in a linear fashion (exch. Mg: quadratic) with the lime rate ($p < 0.001$). Soil pH and BS increased in this layer from 4.41 ± 0.07 and $20.8 \pm 4.2\%$, respectively, for the controls, to

Table 1
Ten-year effect of dolomitic lime rate on forest soil physico-chemical properties

Lime rate (Mg ha^{-1})	pH	Carbon (g kg^{-1})	Organic matter (g kg^{-1})	Kjeldahl N (g kg^{-1})	C/N	Exchangeable cations (cmol kg^{-1}): Ca	Mg	K	Na	Acidity	CEC	BS (%)
0–12 cm depth, including the forest floor												
0	4.10	171	324	7.68	22.4	3.32	0.69	0.45	0.04	6.18	13.55	41.9
2	4.50	151	302	7.19	21.5	4.77	2.08	0.45	0.04	3.72	13.62	62.1
5	4.52	163	315	7.01	23.4	6.19	3.54	0.40	0.04	2.20	16.08	76.0
10	4.93	171	318	7.32	23.7	9.85	7.30	0.40	0.04	1.18	21.26	90.6
20	5.51	106	204	4.77	22.4	9.73	8.14	0.25	0.03	0.30	19.40	97.5
MSE[a]	0.1146	0.1731	0.1537	0.1518	10.07	0.418	0.473	0.159	0.096	0.418	39.94	335
$p > F$[b]	<0.001L	0.008Q	0.003Q	0.003Q	0.668	0.014Q	<0.001Q	<0.001Q	0.018Q	<0.001L	0.005Q	<0.001Q
12–20 cm depth												
0	4.41	95	193	4.30	22.3	0.74	0.20	0.17	0.03	5.47	8.18	20.8
2	4.69	78	159	3.52	22.3	0.86	0.42	0.14	0.03	4.33	6.48	27.7
5	4.50	101	190	4.31	23.8	1.32	0.85	0.15	0.03	5.93	8.97	30.6
10	4.78	97	188	4.52	21.5	1.54	1.32	0.15	0.03	4.47	8.14	41.7
20	4.90	79	156	3.80	21.1	1.96	2.21	0.14	0.02	2.78	8.47	56.8
MSE[a]	0.0623	0.1274	0.1156	0.1274	6.74	0.453	0.416	0.136	0.127	0.246	7.09	229
$p > F$[b]	<0.001L	0.423	0.301	0.841	0.072	<0.001L	<0.001Q	0.412	0.021Q	0.001Q	0.376	<0.001L
20–40 cm depth												
0	4.83	73	147	2.93	24.9	0.26	0.06	0.07	0.03	3.89	4.58	10.9
2	4.97	63	133	2.65	23.9	0.27	0.10	0.05	0.02	2.95	3.90	15.2
5	4.87	75	159	2.98	25.4	0.27	0.12	0.06	0.03	4.16	4.83	10.8
10	4.97	61	130	2.56	23.7	0.27	0.16	0.05	0.02	3.16	3.95	15.4
20	4.95	60	121	2.70	22.3	0.32	0.23	0.05	0.02	2.94	3.81	19.1
MSE[a]	0.0239	0.0957	0.0778	0.0772	7.64	0.522	0.462	0.178	0.089	0.116	1.86	100
$p > F$[b]	0.151	0.119	0.062Q	0.500	0.012L	0.428	<0.001Q	0.200	0.004Q	0.086	0.191	0.050L

[a] Model variance in the original units (natural logarithm for organic carbon, organic matter and total N concentrations and exchangeable Ca, Mg, K, Na and acidity; standard linear for the other variables).
[b] L: linear effect; Q: quadratic effect.

Table 2
Effect of dolomitic lime rate on organic matter, carbon, bulk density and element content in the first 12 cm of soil (includes the forest floor)

Lime rate	Carbon	Organic matter	Kjeldahl N	Bulk density	Exchangeable cations (mol(+) m^{-2})				
(Mg ha^{-1})	(g m^{-2})	(g m^{-2})	(g m^{-2})	(g cm^{-3})	Ca	Mg	K	Na	Acidity
0	65.5	123.2	2.92	0.348	1.49	0.27	0.172	0.016	2.33
2	61.3	121.7	2.87	0.337	2.17	0.97	0.182	0.017	1.47
5	64.7	124.0	2.77	0.345	2.79	1.84	0.156	0.016	0.85
10	67.2	123.9	2.84	0.339	3.87	3.03	0.154	0.015	0.44
20	54.9	104.6	2.46	0.433	4.98	4.27	0.129	0.016	0.15
MSE[a]	216.8	118.5	0.306	0.0127	1.120	0.935	0.002	1.6E-5	0.547
$p > F$[b]	0.472	0.342	0.233	0.432	<0.001L	<0.001L	0.001L	0.972	<0.001L

[a] Model variance in the original units (natural logarithm for exchangeable acidity; standard linear for the other variables).
[b] L: linear effect; Q: quadratic effect.

4.90 ± 0.07 and $56.8 \pm 4.4\%$, respectively, for the 20 t ha^{-1} lime treatment. On the contrary, concentrations of exchangeable Na and acidity decreased in a quadratic fashion with the lime rate in this soil layer, mainly because of the 20 t ha^{-1} lime rate ($p \leq 0.021$). Again, the C/N ratio was not greatly influenced by lime treatments in this soil layer ($p = 0.072$). However, in contrast to the upper soil layer, total carbon, total N and organic matter concentrations were not influenced by the lime treatments ($p \geq 0.301$).

In the 20–40 cm soil depth layer, only exchangeable Mg concentrations and BS increased with the lime rate ($p \leq 0.05$), while exchangeable Na and acidity concentrations decreased with the lime rate, but only marginally in terms of probability and values (Table 1). The increases between the control and the 20 t ha^{-1} treatment represented 74 and 43% for the exch. Mg and BS, respectively. Concentrations of organic matter, C/N ratio and exchangeable acidity decreased with the 20 t ha^{-1} lime rate at marginal probabilities in this soil layer by 18, 10 and 24%, respectively ($p \leq 0.086$). Soil pH, total carbon, total N, and exchangeable Ca, K and CEC concentrations were not influenced by the lime treatments in this soil layer ($p \geq 0.119$).

3.2. Regeneration in the four sub-plots

A total of 832 regeneration stems were numbered and measured in study plots. Stems were distributed, in order of importance, among SM ($n = 689$), AB ($n = 46$), red maple (*Acer rubrum* L.: $n = 29$), striped maple (*Acer pensylvanicum* L.: $n = 23$), hobblebush (*Viburnum alnifolium* Marsh.: $n = 19$), yellow birch (*B. alleghaniensis* Britton: $n = 15$), mountain maple (*Acer spicatum* Lam.: $n = 10$), and Canada yew (*Taxus canadensis* Marsh.: $n = 1$).

Mean regeneration stem density and proportion of SM, AB, other species, and all species pooled together for each lime treatment are presented in Fig. 1. Eleven years after treatment, overall stem density was not affected by lime rate ($p = 0.245$). Nevertheless, lime rate linearly influenced the proportion of stems of SM, AB, and other species pooled together ($p < 0.001$). The proportion of SM over all other species increased from 60% in control plots to around 80% in the 2, 5, and 10 t ha^{-1} lime treatments, and to more than 95% in the 20 t ha^{-1} lime treatment (Fig. 1). Contrarily, the proportion of AB and other species decreased with the lime rate.

The effect of the lime treatment on regeneration height and diameter of SM, AB, other species, and all species pooled together are presented in Fig. 2. On average, regeneration height differed significantly among species ($p < 0.001$), AB being the tallest species (56.6 ± 8.8 cm) followed by other species (36.0 ± 6.1 cm) and SM (22.5 ± 1.8 cm). Eleven years after lime treatments, no effect on regeneration height of any species was detected ($p \geq 0.399$). Height of SM regeneration followed a linear trend, but variability among treatments was too high to detect any significant trend ($p \geq 0.173$). In contrast, SM regeneration diameter significantly increased with lime rate ($p = 0.029$), with the diameter of SM in the 20 t ha^{-1} lime treatment being 33–38% higher than in other treatments. According to the response of regeneration height, diameter of AB and other species stayed more or less constant, independently of lime rate.

3.3. The three taller regeneration stems in the 5 m radius plots

Overall, AB accounted for 73% of the three tallest regeneration stems recorded, while SM and other species each represented 13.5%. Sugar maple was part of the three taller regeneration stems only in the two highest lime rates. The effect of lime treatment on height and diameter of the tallest regeneration stem for SM, AB, yellow birch and all species pooled together are presented in Fig. 3. A linear decreasing trend according to lime rate was detected on height and diameter of the tallest regeneration stems for all species confounded ($p \leq 0.002$) and for AB stems ($p < 0.001$). On average, the height and diameter of the tallest AB regeneration stems decreased by 53 and 50%, respectively, in the 20 t ha^{-1} lime treatment, as compared with the control treatment. Height and diameter of yellow birch also followed a linear trend, but variability among treatments was too high to make it significant ($p \geq 0.087$).

3.4. Crown dieback and light environment

The lime rates influenced both the crown dieback and the light environment under the canopy ($p \leq 0.008$; Fig. 4). Mean crown dieback varied between 0 and 3.4% for lime treatments as compared to 18.4% for the unlimed controls. These results are in accordance with the inverse pattern of light interception by the canopy. Light interception was evaluated to 81% for control trees and was, on average, up to 95% for limed trees.

4. Discussion

4.1. Liming effect on soil

After 10 years, liming drastically changed the properties of this soil, particularly for the 20 t ha^{-1} lime rate, with effects decreasing downward in the soil profile. Liming increased soil pH, exchangeable Ca and Mg, CEC and BS in the first 20 cm, and its effect was also apparent down to 40 cm. Many of the liming effects on the soil in this experiment were already observable after 5 years (Houle et al., 2002). Magnesium appears to have moved downward in the soil profile more rapidly than Ca (Table 1). Similar soil response to liming has been observed in northern hardwood forests 1 year after lime application down to the first 15 cm of soil (Demchik and Sharpe, 2001), from the first to the 7th year in northern hardwoods of north central Pennsylvania (Long et al., 1997), in the Adirondacks, New York (Blette and Newton, 1996), and in European forests (Kreutzer, 1995).

The decrease in exchangeable K and Na concentrations in the soil down to 40 cm and in K content in the first 12 cm may be

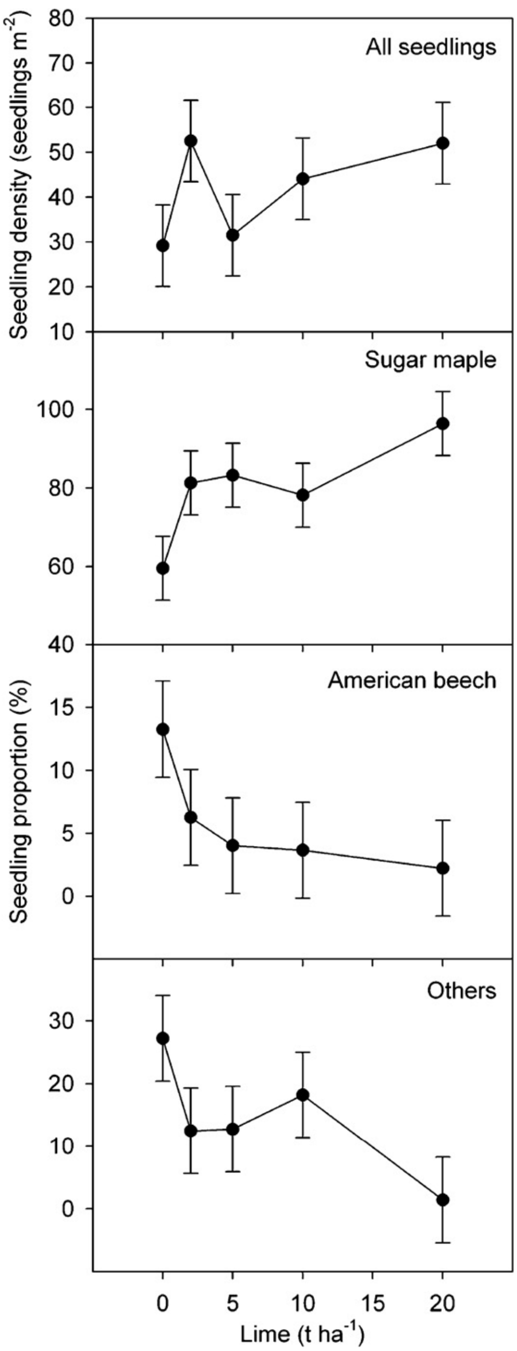

Fig. 1. Mean seedling density and proportions of sugar maple, American beech, other species, and all species pooled together in function of the lime rate. Error bars represent ± standard error.

related to Mg and Ca displacing K ions from soil exchange sites to soil solution, and subsequent leaching (Côté et al., 1995; Ludwig et al., 2002). Reductions in exchangeable K in forest soil after liming have been observed also by Long et al. (1997). This may explain why K concentrations decreased in SM foliage with the lime rate in the first 4 years and up to the 10th year in this experiment (Moore et al., 2000; Moore and Ouimet, 2006) and elsewhere (Burke and Raynal, 1998).

Ten years after liming, total carbon, total N and organic matter concentrations decreased only with the 20 t ha^{-1} lime rate in the first 12 cm of soil (Table 1), containing mainly the forest floor. Along with the decrease in carbon, organic matter, and N content and increase in bulk density (Table 2), the changes in this soil layer suggest that the 20 t ha^{-1} lime rate significantly increased organic matter mineralization (Nodar et al., 1992; Kreutzer, 1995), compared to the lower lime rates. It is hypothesized that the higher mineralization in the higher lime rate reduced the depth of the forest floor, resulting in sampling more mineral soil, thus increasing the bulk density of this sampled soil layer. Seven years after applying lime at a rate of 4 t ha^{-1} in a 80-year-old Norway spruce (*Picea abies* L. Karst.) stand, Kreutzer (1995) observed a decrease in organic C stocks of 23% in the forest floor and an increase of 4% in the first mineral horizon. He attributed this change in organic C to greater earthworm activity. Since earthworms were absent in our experimental site, it appears that soil mixing by these organisms was not a possibility. Also, no other ground dwelling organism, that could be involved in significant soil mixing, was found during this study.

4.2. Liming effect on SM regeneration

In this study, lime application improved the soil acid–base status (Table 1). Also, from this experiment it has been demonstrated that liming improved nutrition, vigor and growth of limed SM trees for at least 10 years after treatment (Moore et al., 2000; Moore and Ouimet, 2006). The same mechanisms seem to have been beneficial for SM regeneration installation and development. The observed increase in the proportion (Fig. 1) and mean diameter (Fig. 2) of SM regeneration stems, 11 years following lime application, are in accordance with other studies that reported higher abundance and higher relative diameter of SM seedlings in Ca-treated areas (Long et al., 1999; Kobe et al., 2002; Juice et al., 2006). Moreover, a fertilization study in Connecticut suggested that Ca is an important growth-limiting nutrient in juvenile phase of the SM life cycle in northern hardwood forests of eastern North America (Bigelow and Canham, 2007).

It has been previously suggested that soil calcium may be a limiting factor in SM seedling survival in forests with low soil calcium (Jenkins, 1997; Long et al., 1999; Duchesne et al., 2005; St-Clair and Lynch, 2005; Juice et al., 2006; Bigelow and Canham, 2007). There is evidence that the distribution of SM is related to the availability of soil mineral nutrients (Whitney, 1991; Van Breemen et al., 1997). Growth of SM seedlings is directly proportional to soil base saturation (Ouimet et al., 1996). Its survivorship has also been correlated with soil pH and, presumably, nutrient availability in northwestern Connecticut and central-western Michigan (Kobe et al., 1995). In contrast, the relative dominance of AB in the canopy has been inversely related to soil pH and calcium availability (Arii and Lechowicz, 2002), such conditions arising from soil acidification. Moreover, soil acidification also increases Al and Mn mobilisation. High Al and Mn concentrations may be toxic or may even limit Ca and Mg uptake by SM seedlings (Thornton et al., 1986; McQuattie and Schier, 2000). Poor soil quality due to low soil nutrient availability and Al toxicity were deemed as among the causal factors of SM decline in Quebec (Ouimet and Camiré, 1995; Moore et al., 2000; Duchesne et al., 2002; Moore and Ouimet, 2006). From all these studies we can conclude that liming of acid soils in northern hardwood stands appears to favor SM regeneration. However, long-term studies are needed in order to better understand the effect of liming on regeneration and its consecutive impact on structure and composition of northern hardwood stands.

4.3. Combined effect of light and soil nutrient availability on regeneration

Availability of light and soil resources are two important factors governing composition and development of the regeneration. Sugar maple and AB have a distinctive response to those parameters. Sugar maple is generally recognised as being less

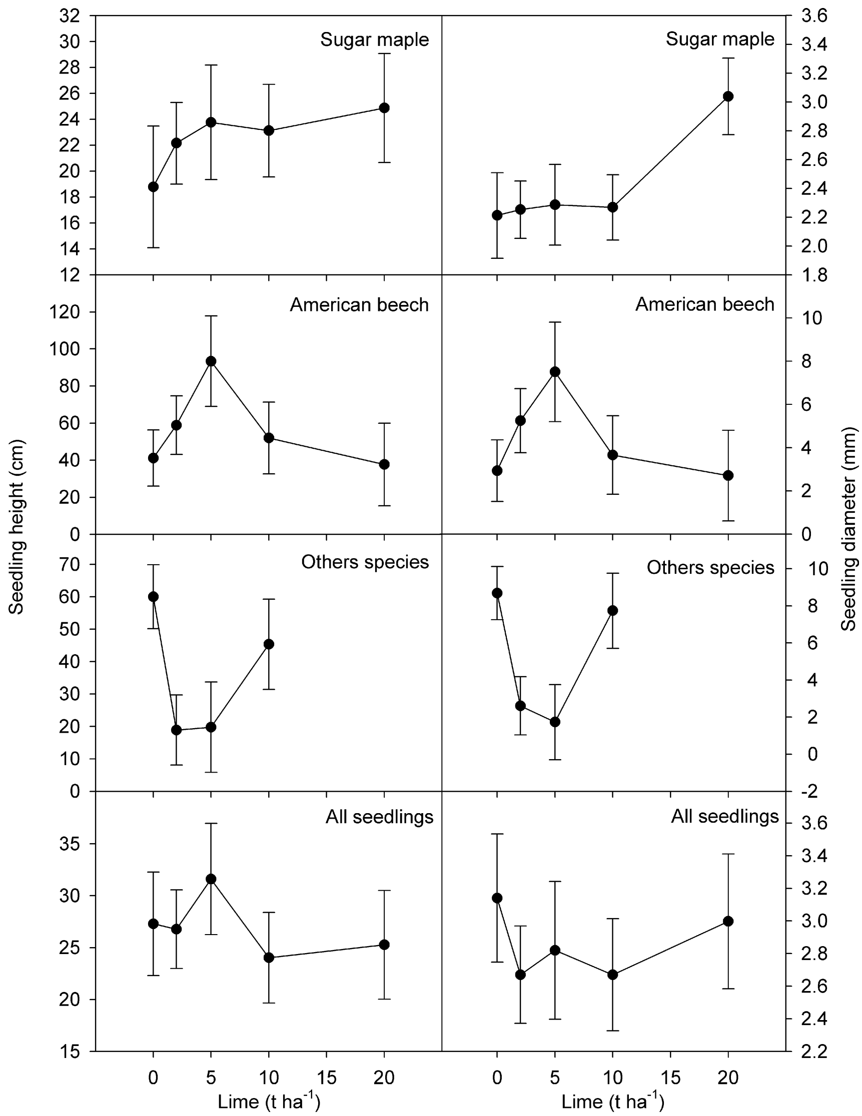

Fig. 2. Mean seedling height and diameter of sugar maple, American beech, other species, and all species pooled together. Error bars represent ± standard error.

shade tolerant than AB. Mortality of SM is higher than AB under very shady conditions (Canham, 1988; Pacala et al., 1994; Kobe et al., 1995; Poulson and Platt, 1996). Sugar maple is also recognised as being more sensitive to the acid–base status of soil as compared to AB (Long et al., 1997). In our study, the addition of lime improved the acid–base status of soil (Table 1). Concurrently, the lime treatment improved the vigor of the overstory trees, reducing the light availability in the understory (Fig. 4). These two major processes acted together to govern the ecological response of regeneration to lime treatments. However, given that the response pattern of SM regeneration stems in terms of stem diameter and species proportion (Fig. 2) was not similar to those of crown dieback and light interception (Fig. 4), but more in line with lime rate and soil response (Table 1), we suggest that SM regeneration improvement with lime rate was mainly linked to an improvement in soil fertility rather than to the light environment.

On average, height and diameter of the three tallest regeneration stems were inversely correlated to the lime rate 11 years following liming (Fig. 3). This could have been the result of (1) an increase in the stem abundance with lime rate, increasing competition for space and soil resources and limiting the development of the taller regeneration, (2) the higher canopy closure under limed trees that could have limited the development of the regeneration (Fig. 4), or (3) a negative effect of lime on regeneration development. Our results do not support the first hypothesis because no increase in regeneration density was detected with lime rate (Fig. 1). Comparison of patterns of the taller AB regeneration response in the understory (linear response: Fig. 3), with those of crown dieback, light interception (quadratic and cubic response: Fig. 4), and soil (linear response: Table 2) suggests that, although canopy closure under limed trees may have contributed to reduction of the taller AB regeneration development, it was not the main cause explaining this decreasing trend. Consequently, it appears that the change in soil fertility was the main factor explaining the responses of the taller regeneration (mainly AB) in the understory (Fig. 3), as previously suggested for SM regeneration proportion and diameter.

It should be noted that the tallest regeneration stems were mainly AB (73%), given that AB was predominant in the understory when the lower vegetation was removed (see Section 2.2), and that AB has the ability to reproduce itself vegetatively by sprouting, thus favoring AB early development following liming. However, SM stems were observed among the taller regeneration only in the two higher liming rate treatments. This result seems quite meaningful given that (1) AB is sprouting more aggressively than

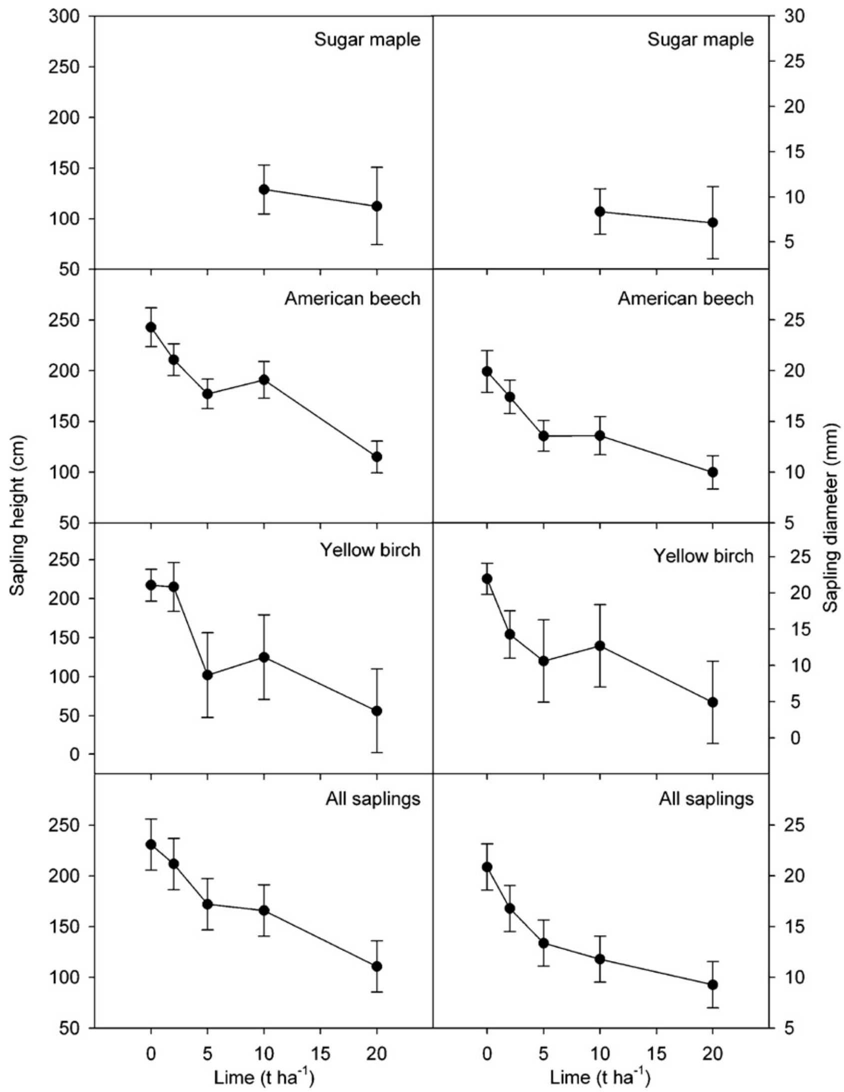

Fig. 3. Mean height and diameter of the taller regeneration stems of sugar maple, American beech, yellow birch, and all species pooled together. Error bars represent ± standard error.

SM and (2) AB is growing better than SM under shady condition. This suggests, in the conditions prevailing for the two higher liming levels, that SM may have been vigorous enough to compete with the taller AB regeneration stems in the understory.

Although some studies have proposed other factors to explain the recent increase in the number of AB poles in some SM stands in northeastern USA (see Section 4.4.3), our results and those of Duchesne et al. (2005) suggest that, in Quebec, this phenomenon is not a consequence of the ability of AB to perform under the deep shade of old growth, but rather the result of a fast colonization by an opportunistic species that is better adapted to site conditions following changes in soil properties, and to SM decline and dieback. Base cation availability, being generally lower and less adequate for SM regeneration development in declining stands, favored AB expansion (Fig. 2: stem %; Duchesne et al., 2005). However, interactions among soil, light and regeneration growth and abundance remain to be established experimentally.

4.4. Possible confounding factors

4.4.1. Delay in soil and regeneration responses caused by lime dissolution

In this study, many soil responses seem to follow a step function at the higher lime dose (20 t ha^{-1}), suggesting that, in the low doses, there would be a prolonged delay until sufficient lime dissolved to influence the soil and vegetation. Hence, seedling responses would be delayed by both the delayed soil chemical and canopy recovery responses. However, some studies on lime dissolution do not corroborate this hypothesis while others do. These studies show that lime dissolution rate is not a function of rate of application, but mainly of soil pH and lime particle size (Wallin and Bjerle, 1989; Musil and Pavlicek, 2002). Fine powdered dolomitic lime, as used in this study, should have quickly reacted in such acid forest soil (within 5 months according to Swartzendruber and Barber (1965), but 3 years according to Musil and Pavlicek (2002), and up to 6 years according to Kreutzer (1995)). However, an indication of the quick lime reaction of soils in this liming experiment was the early SM tree response with increased foliar Ca and Mg concentrations and increased growth 1 year after lime application, even at the lowest lime dose (Moore et al., 2000). Therefore, these last results suggest that possible delay in soil and regeneration responses caused by the time of lime dissolution appears negligible.

4.4.2. Higher SM seed production after liming

In a previous study, Long et al. (1997) reported higher SM seed production after liming in Pennsylvania, but liming had no effect on seed crop frequency. In our study, no attempt was made to evaluate SM seed production following liming. In this context, it is

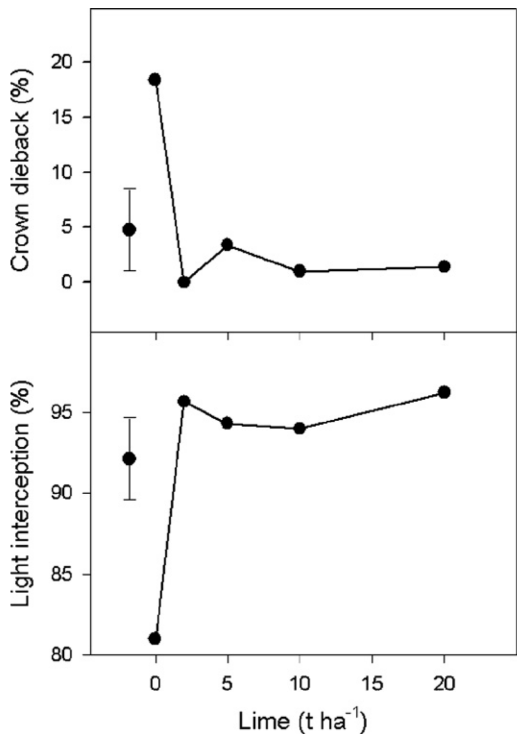

Fig. 4. Mean crown dieback and light interception of mature sugar maple trees in function of the lime rate.

likely that the increased SM vigor following liming in our study (Moore and Ouimet, 2006) could have been beneficial for seed production and could explain, in part, the higher abundance and proportion of SM regeneration stems in limed plots. However, the experimental plot design in our study (5-m radius around each SM tree) is quite different from the 45 m × 45 m plots set up by Long et al. (1997). While some studies reported that mean dispersal distance of SM seedlings from the parent tree varies from 8 to 10 m (Ribbens et al., 1994; Caspersen and Saprunoff, 2005), others showed that seed dispersal into an harvested area declined exponentially with distance from the forest edge, but the number of seeds was approximately 50 m^{-2} at 25 m and 25 m^{-2} at 50 and 75 m (Hughes and Fahey, 1988). From these studies and our experimental plot design, we can conclude that the seeds falling under single SM trees in our study also came from around other SM trees and, consequently, the possibility of a strong confounding effect of SM seed crops and soil amelioration on SM regeneration seems limited.

4.4.3. Other mechanisms

Some other mechanisms have been proposed to explain AB and SM regeneration dynamics in some particular stands of the northern hardwood forest, notably: (1) differences in shade tolerance of both species (Canham, 1988; Beaudet et al., 1999); (2) differences in the reproductive strategy of both species (Jones and Raynal, 1988; Hane, 2003); (3) changes in soil properties resulting from an increased AB density in forest stands, which may be detrimental to SM regeneration (Dijkstra et al., 2001); (4) phytotoxicity of AB leaf leachate to SM regeneration (Hane et al., 2003); (5) herbivore preference for SM in comparison to AB (Marquis and Brennemann, 1981); and (6) land-use history (Siccama, 1971). However, many of these mechanisms were not involved in this study (no observation of herbivory, no major perturbation, no significant extreme climatic event, no forest harvesting since 60 years). Moreover, if present, few of these factors could have contributed to confounding the effect of liming on SM and AB regeneration in our study given that most of them would have affected both control and treated plots.

4.5. Implications for forest management

Many SM stands in Quebec are undergoing changes in their structure and composition, given regeneration failure of SM and expansion of AB (Duchesne et al., 2005). These changes represent serious challenges for the sustainability of the SM population in many areas of the province. This phenomenon is a preoccupying issue for forest managers, given that SM is much more valuable than AB. Many field trials have been carried out in attempt to control the AB expansion in SM stands, including clearing the understory before selection cutting, varying the harvesting intensity, and applying herbicide treatments. In New York, chemical herbicides have demonstrated their effectiveness in controlling the understory vegetation (Ray et al., 1999). However, in 1994, the Quebec government committed itself to end the use of chemical herbicides in forests in its *Stratégie de protection des forêts* (Forest Protection Strategy). In this context, liming offers a promising tool to limit AB expansion and favor SM establishment and development. Our results show that even with the lowest application levels, liming could increase the proportion of SM regeneration stems in the understory of base-poor and declining SM stands. However, even if Ca addition could positively influence SM regeneration dynamics (Fig. 1; Long et al., 1997; Kobe et al., 2002; Juice et al., 2006) and mature SM trees (Wilmot et al., 1996; Long et al., 1997; Moore and Ouimet, 2006), AB regeneration needs to be mechanically controlled following lime application to ensure the free development of the SM regeneration. The results suggest that liming, coupled with mechanical vegetation control, should favor the development of SM regeneration and limit the expansion of AB in base-poor and declining SM stands. A study dealing with this topic is underway near the liming experimental site, and the results should be available in the next few years. Finally, to prevent damage to existing root systems and to limit root sprouting of AB, aerial lime application should be the method used when possible.

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